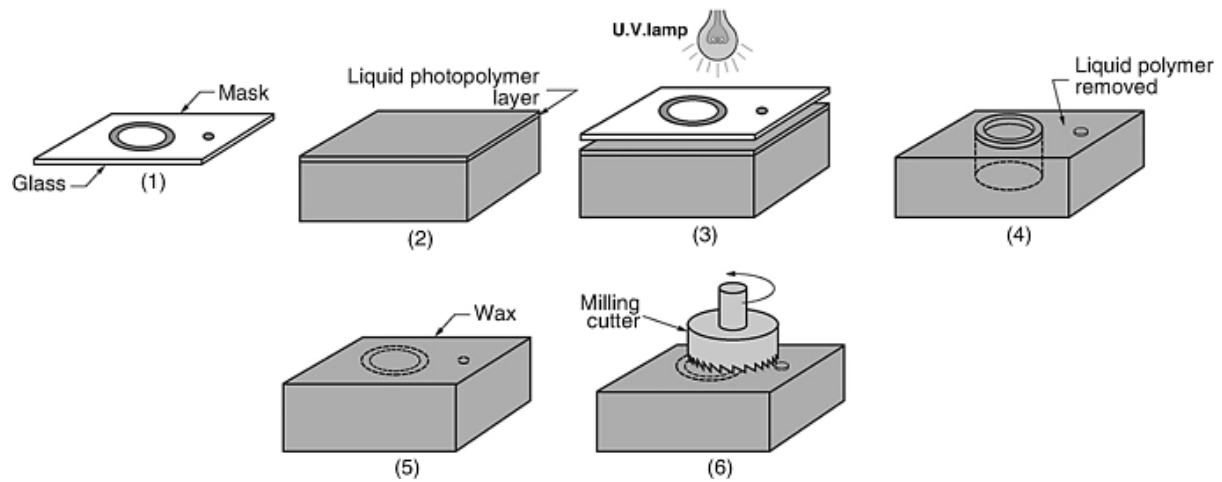


SOLID GROUND CURING



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2.3.1 INTRODUCTION

Solid Ground Curing (SGC) is a resin-based rapid prototyping process manufactured by Cubital in Israel. SGC materials are generally UV-curable photopolymers, similar to those used in stereolithography, but formulated for high-speed, full-layer exposure. These materials are typically acrylate or epoxy-based resins with additives that enhance curing speed, dimensional stability, and mechanical strength. Some formulations may include fillers or pigments to improve surface finish and reduce shrinkage. The solidified resin layers are supported by wax material, which fills the non-cured regions during the process and is later melted away, resulting in parts with excellent dimensional accuracy, smooth surface finish, and minimal post-processing requirements.

2.3.2 MACHINE DETAILS

The SGC systems are physically quite large, set up in an automatic assembly line fashion with the curing lamp, resin vacuum, milling station, etc. They are currently offered in two systems, the entry level SGC 4600 and the high-end SGC 5600. The SGC 4600 systems offers a 350mm x 350 mm x 350mm (14" x 14" x 14") working envelope, whereas the SGC 5600 system has a 500mm x 350mm x 500mm (20" x 14" x 20") envelope and a 300% increase in net production rate over the SGC 4600 system.

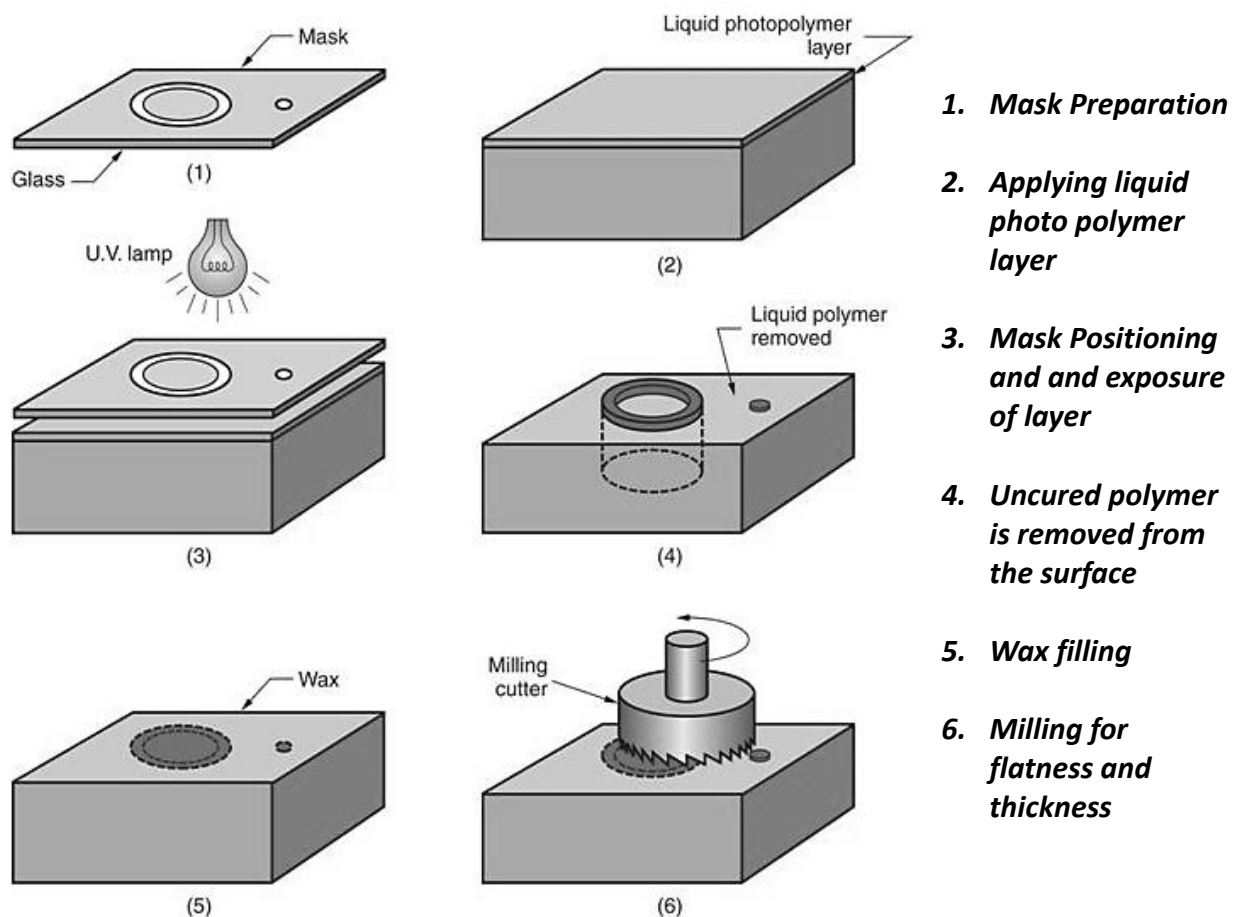
Specifications of SGC Machines

Model	Solider 4600	Solider 5600
Irradiation medium	High power UV lamp	
XY resolution (mm)	Better than 0.1	
Surface definition (mm)	0.15	0.15
Elevator vertical resolution (mm)	0.15	0.1–0.2
Minimum feature size (mm)	0.4 (horizontal, X–Y) 0.15 (vertical, Z)	0.4 (horizontal, X–Y) 0.15 (vertical, Z)
Work volume, XYZ (mm × mm × mm)	350 × 350 × 350	500 × 350 × 500
Production rate (cm ³ /hr)	550	1311
Minimum layer thickness (mm)	0.06	0.06
Dimensional accuracy	0.1%	0.1%
Size of unit, XYZ (m × m × m)	1.8 × 4.2 × 2.9	1.8 × 4.2 × 2.9
Data control unit	Data Front End (DFE) workstation	
Power supply	380–415 V _{AC} , 3 phase, 50 kW	380–415 V _{AC} , 3 phase, 50 kW

2.3.3 PRINCIPLE

SGC process is based on the principle of layer-by-layer photopolymer solidification using a high-intensity ultraviolet (UV) light source combined with a mask projection system for full-layer exposure. The main principles of the process are as follows:

1. Parts are built layer by layer from a liquid photopolymer resin that solidifies when exposed to UV light. Unlike stereolithography, which uses a laser to trace each cross-section, SGC employs masked illumination, where the entire layer is exposed simultaneously. The image of each layer is generated by a non-impact ionographic printing process, similar to the xerographic process used in photocopiers.
2. Multiple parts can be processed simultaneously by grouping them into batches. This capability enhances productivity by enabling parallel fabrication of several components within the same build cycle.
3. Each layer of the build contains cross-sectional slices of one or more parts. All slices in a single layer are created simultaneously, ensuring high throughput. Layers are intentionally formed slightly thicker than required, and after curing, a milling operation is performed to achieve precise layer thickness and excellent vertical accuracy.
4. The SGC process is self-supporting, eliminating the need for additional support structures. The unexposed regions of each layer are filled with meltable wax, which solidifies and provides mechanical support during fabrication. Once the build is complete, the wax is melted or dissolved away, revealing the finished part with smooth surfaces and accurate dimensions.



Principle of SGC

PROCESS

The SGC process begins with the preparation of a 3D CAD model of the part to be produced. The model is digitally sliced into thin cross-sectional layers using Cubital's Solider

DFE (Data Front End) software, which verifies, corrects, and converts the CAD file into an acceptable format such as STL. The processed slice data are then transferred to the mask generation unit, where an electrostatic charge image of each layer is created on a transparent mask plate through a non-impact ionographic process, similar to xerographic printing. The charged image is developed using black electrostatic toner to form an opaque pattern that represents the solid regions of the part.

Once the mask is generated, a thin layer of liquid photopolymer resin is spread uniformly over the build surface. The mask is positioned above the resin layer, and the entire surface is exposed to a collimated UV light source, curing the exposed regions corresponding to the layer geometry. The unexposed resin remains in liquid form and is removed by vacuum suction. The resulting voids are then filled with molten wax, which solidifies upon cooling to provide complete mechanical support. The layer is then milled to an exact thickness, producing a flat, smooth surface ready for the next layer.

This sequence of operations resin coating, masked UV exposure, resin removal, wax filling, cooling, and milling is repeated automatically until the final layer is completed. In advanced SGC systems such as the Cubital Solider 5600, the part undergoes an additional UV flood-curing step after each layer to ensure complete polymerization.

Once all layers are finished, the supporting wax is melted or dissolved away, leaving behind a fully solid, dimensionally accurate, and smooth-surfaced prototype.

ADVANTAGES

1. **Parallel processing:** The process is based on instant, simultaneous curing of a whole cross-sectional layer area (rather than point-by-point curing). It has a high speed throughput. Many parts can be produced at the same time.
2. **Self-supporting:** The solid wax supports the part in all dimensions and therefore a support structure is not required.
3. **Job changing is easier:** Removable trays allow job changing during a run and layers are erasable.
4. **Unique part properties:** The part that the Solider system produces is reliable, accurate, sturdy, machinable, and can be mechanically finished.
5. **Minimum shrinkage effect:** This is due to the full curing of every layer.
6. **High structural strength and stability:** This is due to the curing process that minimizes the development of internal stresses in the structure. As a result, they are much less brittle.
7. **No hazardous odors are generated:** The resin stays in a liquid state for a very short time, and the uncured liquid is wiped off immediately. Thus, safety is considerably higher.

DISADVANTAGES

1. **Requires large physical space:** The size of the system is much larger than other systems with a similar build volume size.
2. **Wax gets stuck in corners and crevices:** It is difficult to remove wax from parts with intricate geometry. Thus, some wax may be left behind.
3. **Waste material produced:** The milling process creates shavings, which have to be cleaned from the machine.

4. **Noisy:** The Solider system generates a high level of noise as compared to other systems.

2.3.4 APPLICATIONS

The SGC process became obsolete due to its high system complexity, expensive equipment, and slow overall build rate. Although it could cure entire layers simultaneously, the additional steps such as waxing, cooling, and milling made it time-consuming and costly to operate. The process also required frequent maintenance, occupied large space, and consumed significant energy. With the emergence of simpler, faster, and more economical technologies like SLA, SLS, FDM, and PolyJet printing, SGC gradually lost its relevance and is discontinued.

However, following are some of the applications

1. **Functional Prototypes:** Used to create durable and highly accurate prototypes for functional testing in industries such as automotive, aerospace, and consumer products.
2. **Tooling and Molds:** Ideal for producing precise molds, patterns, and tooling inserts due to its excellent dimensional accuracy and superior surface finish.
3. **Casting Patterns:** Commonly applied in manufacturing investment casting and sand casting patterns where fine details and smooth surfaces are essential.
4. **Complex Geometries:** Suitable for fabricating parts with intricate internal structures, undercuts, and overhangs, effectively supported by wax during the build process.